SHAIKH ABDUL ZUBAIR ABDUL KABIR

Full-Stack & Mobile Developer | Edge AI & Offline Mesh Systems Specialist

📍 India 📧 abdulzubair221@gmail.com 🌐 github.com/abdulzubair1135

🎓 BCA Final Year — SRK Institute (Graduating 2026/2027)

PROFESSIONAL SUMMARY

Prolific and versatile **Software Developer & BCA Final Year Student** at **SRK Institute of Business Administration and Computer Education** with **55+ completed projects** across Android Native, Flutter ecosystems, Full-Stack web platforms, Local Edge AI, and Game Engines. Completed a 50-day **Software Engineering Internship (83.7%+ productive rating)**. Deep expertise in offline-first architectures, peer-to-peer mesh networking (BLE/Wi-Fi Direct), cryptographic validation (Ed25519), and browser automation agents.

CORE TECHNICAL SKILLS

Languages	Java, Kotlin, Dart, Python, TypeScript, JavaScript, SQL, C#, C++, GDScript, C, HTML5/CSS3
Mobile & Cross-Platform	Android Native (Java/Kotlin, Jetpack Compose, Coroutines, AccessibilityService), Flutter (Dart, Multi-App, SQLite), React Native
Web & Backend	Node.js, Express.js, React 18, Vite, Tailwind CSS, REST APIs, WebSockets, Socket.IO, Leaflet.js, Monaco Editor, xterm.js
AI & Edge Computing	Local LLMs (Llama, Ollama), Android On-Device Inference, PyTorch Custom Neural Networks, Agentic Loops (RAG), Playwright Automation
Databases & Cloud	Supabase (PostgreSQL), MongoDB, SQLite, Firebase (Firestore, Cloud Messaging FCM, Auth), Render Cloud
Networking & Security	BLE (Bluetooth Low Energy), Wi-Fi Direct, P2P Mesh Routing, Ed25519 Signatures, Dynamic Nonce QR, PathGuard Sandbox
Tools & Game Engines	Android Studio, VS Code, Visual Studio, Git/GitHub, Postman, Unity 3D (NavMesh), Godot Engine 4, Arduino IDE, FFmpeg

WORK EXPERIENCE & INTERNSHIP

Software Developer Intern — Full-Stack & Engineering **50 Calendar Days (83.72% Productive Efficiency Rate)**

- Contributed to end-to-end software module development, API integration, and architectural debugging across distributed codebases.
- Engineered offline caching, database schema improvements, and frontend performance optimizations.
- Collaborated across agile remote sprints with structured daily tracking, issue resolution, and system reporting.

FLAGSHIP ENGINEERING PROJECTS

SRK Campus Connect — Campus Presence & Faculty Coordination System

Flutter (Dart), Node.js, TypeScript, Express, MongoDB, Socket.IO, Ed25519

- Architected an **offline-first campus ecosystem** unifying 3 Flutter client apps (Student App, Faculty Dashboard, Admin Console) and a unified Node.js/TypeScript backend for SRK Institute.
- Engineered **dynamic 15-second rolling QR codes** with rotating nonces and Ed25519 cryptographic signatures to eliminate proxy attendance and replay attacks.
- Implemented an autonomous **P2P peer-forwarding gateway election** protocol allowing offline students to securely relay attendance packets through connected peers.
- Built a multi-factor **Presence Confidence Engine** calculating weighted evidence scores (0–100%) using BLE proximity and cryptographic timestamps.

AapdaSetu / NearMeshChat — Disaster Response & Offline Mesh Network

Android Native (Java), BLE Radar, Local Llama Edge AI, React.js, Express, Leaflet

- Engineered an emergency disaster communication network operating completely independent of active cellular towers or internet access.
- Built an **Android client featuring a real-time BLE Radar scanner**, peer distance estimation, and mesh packet store-and-forward routing.
- Integrated **on-device local LLM edge inference (Llama)** directly on Android to provide medical triage, survival guidance, and emergency response instructions without network connectivity.
- Created a centralized **Emergency Operations Command Web Dashboard** with real-time incident mapping using Leaflet and bidirectional Socket.IO dispatch.

Autonomous Web-Agent IDE — Local Natural-Language Software Generator *Python, React 18, TypeScript, Tailwind CSS, Monaco Editor, Playwright, xterm.js*

- Built a local desktop IDE transforming natural-language project requirements into full-stack, tested, and running code using web-based LLMs.
- Created headless browser orchestration with **Playwright to drive web LLMs** without requiring commercial paid API keys.
- Implemented **PathGuard security sandbox** preventing directory traversal and symlink attacks with strict SAFE/CAUTION/BLOCKED policies.
- Integrated multi-tab **Monaco Editor**, live terminal streaming via **xterm.js**, task graph DAG, and automated Git checkpointing with loop-breaking error recovery.

AURA — Autonomous Local AI Research & Learning System *Python, Local LLMs, Vector Memory DB, Playwright, DuckDuckGo API*

- Designed a self-directed cognitive engine implementing a continuous **Think → Plan → Execute → Verify → Reflect → Store** loop.
- Optimized for local consumer hardware (NVIDIA RTX 2050 4GB VRAM) for autonomous knowledge gap discovery using an internal "Curiosity Engine".
- Built a multi-tier memory hierarchy (Semantic, Episodic, Procedural, and Skill Memory) with automated experimentation in sandboxed sub-processes.

ProjectFlow Intelligence — Enterprise Operations Platform *Node.js, Express, React 18, Tailwind CSS, Supabase PostgreSQL, Firebase FCM, Android*

- Enterprise operations platform deployed live on Render Cloud with Supabase PostgreSQL cloud persistence.
- Integrated **Firebase Cloud Messaging (FCM)** for instantaneous push notification routing across web consoles and native Android clients.
- Engineered an offline-first synchronization engine ensuring zero data loss during unstable field network conditions.

Nexus (MeshLinkOffline & PocketAI Server) — Edge AI Microserver *Android Native (Kotlin/Java, 61 source files), Local AI Engine, NanoHTTPD*

- Mobile edge-computing system hosting a local on-device HTTP microserver (`PocketAiHttpServer`) and an embedded AI inference engine.
- Allows nearby devices on the local Wi-Fi/Hotspot network to query AI knowledge bases and retrieve computed answers without active internet connectivity.

DOMAIN PORTFOLIO & ADDITIONAL PROJECTS INVENTORY (55+ TOTAL)

• **Full-Stack & Campus Systems:** MasiPay (Canteen Ordering & Token System), Smart Expenses Tracker (Offline-First Finance), Follow Or Not (Audience Retention Tracker), Social Media Platform (React/Node/MongoDB), Modular MVC Backend Architecture, College Hub Portal, AbdulServer.
• **Offline-First & Android Native:** RecoverX (Forensics-Informed File Carving), Couple Canvas (P2P Collaborative Whiteboard), Emergency Recorder Fast Rec (Compose Panic Utility), Personal Life Vault (Encrypted Storage), Focus Guard (Digital Addiction Blocker), Z-Music Player, myhelper & NearDrop (P2P Wi-Fi Direct), Android Prototypes (MyApplication 1–4).
• **Autonomous AI & Generative Media:** My Custom AI (PyTorch Model from Scratch), AI Zubair LLM Research (Colab Audit Suite), My Helper Agent (RAG Brain), OmniDev AI (Godot Synthesizer), StickForge AI (Stickman Studio), AI Content Repurposing Studio (9:16 Shorts), Auto YT Generator.
• **Game Development (Godot 4 & Unity 3D):** Draw Your Army: WarForge (Tactical RTS), Minecraft 3D Block Engine, Full-Stack RPG Game Suite, Isekai World Adventure, Shadow Kingdom (2D Pixel Art), Unity 3D FPS Game (NavMesh AI), Slikj Water Physics Simulation, Yorus 3D World, 360 Journey Panorama.
• **Embedded Systems & Utilities:** Arduino IoT Wire Communication (I2C), College & School Portals, Precision Stopwatch & Calculator Suite, Systems & Algorithms Repositories (C/C++), AutoHotkey Suite, MongoDB Advanced Aggregations.

EDUCATION

Bachelor of Computer Applications (BCA) — Final Year **Graduating 2026 / 2027**
SRK Institute of Business Administration and Computer Education